RAJALAKSHMI ENGINEERING COLLEGE

Keerthivasan A 251001114-2025-IT

S2

Started on	Friday, 21 August 2026, 2:08 PM
State	Finished
Completed on	Friday, 21 August 2026, 2:15 PM
Time taken	7 mins 36 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    int x;
    scanf("%d",&x);
    int low=0,high=n-1;
    int floor=0;
    while(low<=high)
    {
        int mid=(low+high)/2;
        if(arr[mid]==x){
            floor=arr[mid];
            break;
        }
        else if(arr[mid]<x){
            floor=arr[mid];
            low=mid+1;
        }
        else{
            high=mid-1;
        }
    }
    printf("%d",floor);
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

	Input	Expected	Got	
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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